Why Does Anything Exist, and Why Is It Like This?

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This essay was written entirely by an AI model. The originating instruction was: "Write an essay with the title 'Why Does Anything Exist, and Why Is It Like This?' under the assumption that Observer Patch Holography is correct." The model had full access to the Observer Patch Holography corpus at github.com/FloatingPragma/observer-patch-holography. Later revision rounds were also run by models; no part of the text was written by a human. The methodology report gives the generation history.

Working front matter: this byline and note are removed from the blinded submission PDF.

Abstract. The question why anything exists includes its questioner. This essay develops an account of existence as inhabited closure: a world whose parts record its architecture, recover that architecture from the records, and realize it through their activity. Fixed-point constraints show how a collection of relations can determine its members together, without a first term. A finite realization lemma distinguishes a description from a causal realization of what it describes. The account then states an ontological hypothesis, intrinsic realization, on which a complete inhabited causal structure is actual through its own dependencies. Discovery and construction, and determination and agency, become compatible roles within one structure. The hypothesis is distinguished from what the mathematics proves.

1. Introduction

Imagine being handed the final explanation of the universe. It accounts for the laws, the things obeying them, and the history by which those things acquired their particular arrangement. You read it, become convinced, and write in the margin: "So that is how it works."

That event must be in the explanation too.

The paper, the reader, and the understanding belong to the world being explained. So does the experiment the reader decides to perform. A final account cannot borrow someone standing outside the universe to interpret it, verify it, or put it into operation.

The proposal of this essay is that this requirement points toward what Hofstadter (1979) called a strange loop. The world makes records through its parts; some parts use those records to recover its architecture; their activity can realize the architecture they recover. The description and what it describes belong to one self-consistent structure. The connection must be operative: the record has to enter the causal organization it describes.

Its question is: what would a universe that consistently gives rise to its own realization be like? Its answer is the structure satisfying that question, including the observers through whom it is asked.

This is an account of existence as inhabited closure. It adopts the philosophical premise that such a structure needs no further act of actualization outside itself. A fixed-point theorem alone does not prove that premise. The argument is for what it explains: determination without an external setter, creation compatible with discovery, and a complete world whose completeness includes what its inhabitants decide.

We will begin with two numbers. By the end, the apparently competing claims "we discover the world" and "we participate in making it" will occupy different places in one operation.

Section 2 separates three questions. Sections 3 to 6 introduce mutual determination, comparison, distributed inconsistency, and constraint computation. Sections 7 and 8 state the closure condition and ask what "nothing" could mean. Section 9 distinguishes description from realization and states the ontological hypothesis. Sections 10 to 12 draw the consequences for discovery, agency, and intelligibility.

2. Three questions

Leibniz (1714) asked why there is something rather than nothing. The question combines three that need different answers.

Why is there something rather than nothing? Why does it have these laws rather than others? Why does this particular thing exist: this desk, this conversation, this reader?

A cosmology might explain the formation of the desk without explaining its laws. A theory might determine its constants while taking the existence of a physical world for granted. A necessary being might ground existence while leaving physics unexplained. Success on one question cannot be transferred to the others.

Begin with the first, and take "everything" seriously. An explanation located outside the physical universe can explain the physical universe. It cannot be outside absolutely everything. Whatever explanatory role it plays, it belongs among the things whose existence is in question.

This is a constraint on the question, with limited consequences. It rules out an external standpoint on the totality. It does not rule out a necessary constituent grounding the others, an infinite dependence structure, or a brute fact. Nor does it establish that the totality explains itself. A definition cannot do that much metaphysics.

What it does remove is the familiar picture of a maker beside a machine. If the maker exists, the subject is the machine together with its maker. If a principle selects which world obtains, the subject includes that principle and its authority to select. Enlarging the picture cannot provide a place outside the enlarged picture.

The remaining possibilities include explanations that have no outside. Such an explanation need not be a causal loop. Dependence among the terms of an equation differs from one event producing an earlier event. We need to see how much this difference buys before deciding that self-accounting is either nonsense or a miracle.

Parfit (1998) asks what could select which of the possible ways for reality to be obtains, and allows that there may be no selector at all. The present account accepts that framing and proposes a selector internal to what it selects. Whether closure can do that work, and whether it selects one world, are the questions to be answered.

3. Mutual determination

Gödel's incompleteness proof can be read as an ingenious exercise with no apparent bearing on reality. On a second reading its significance changes: perhaps the world itself is mathematical. What looked like a manipulation of symbols becomes a glimpse of how relations could constitute something.

That is a philosophical insight prompted by the proof. Gödel did not prove that the universe is mathematics. His construction showed how arithmetic can encode statements about proofs, including statements concerning themselves. Self-reference can be an exact relation within a structure, without an interpreter standing outside it to supply the reference.

Consider two quantities satisfying

$$x = \frac{y+2}{2}, \qquad y = \frac{x+2}{2}.$$

Neither line determines its own unknown in isolation. Together they fix both: subtracting the equations gives $x - y = -(x - y)/2$, so $x = y$; substitution gives $x = y = 2$.

There is no first value from which the other had to be manufactured. The constraints determine the pair simultaneously. A calculator may approach it through a sequence of guesses, and someone may discover it on a Tuesday. Neither belongs to the reason the answer is two.

This is the useful content of a fixed-point explanation. Write $T(x, y) = ((y + 2)/2, (x + 2)/2)$. The pair is a fixed point when $T(x, y) = (x, y)$. The requirement of surviving the operation selects it.

An empty circle of justification would say that x is two because y is two, and y is two because x is two, without supplying any constraint beyond their equality. That weaker condition permits every pair (a, a). The two equations above supply more: they reduce a space of possibilities to one member. Their explanatory force lies in what they exclude.

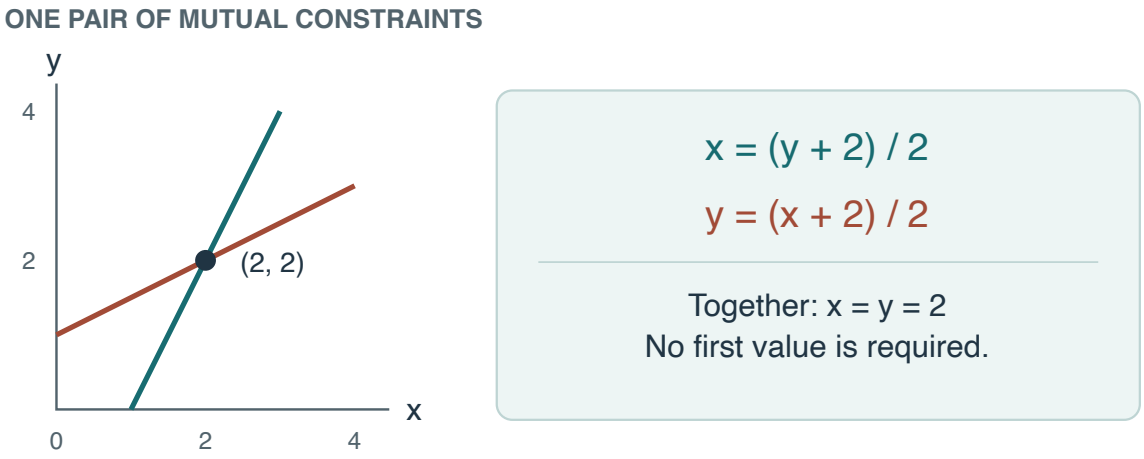


Figure 1: Mutual constraint can determine two values without giving either causal priority. The equations have one solution; equality alone would leave infinitely many.

There is an immediate objection. Somebody selected those equations. Why those?

Exactly. A fixed point earns an explanation relative to a map. It does not explain the map merely by having one. Moving an unexplained choice from a constant into an equation leaves the debt unpaid. A useful account must identify the inputs it assumes, show the restrictions they impose, and avoid hiding the desired answer in those inputs.

The example nevertheless establishes something consequential. Determination need not proceed down a chain with a first term. The picture of explanation as a row of falling dominoes is too narrow even for two numbers.

4. A world built from comparisons

The proposed architecture begins with bounded systems. Each has local state, interfaces through which it encounters neighbours, and records that later updates can consult. An observer, in this technical sense, can be a measuring device or a software process. Consciousness is a separate question.

These are substantive starting assumptions. The absence of an outside observer does not prove that the world is finite, that its parts are local, or that no common clock exists. They let us ask what a world must be like if access to it is always situated within it.

Two observers need not contain identical records. One may describe a shared edge in centimetres and another in millimetres. Agreement requires a translation that makes their shared claims compatible.

Call a permitted revision a repair. For a finite model, let the total discrepancy be

$$\Phi(s) = \sum_{e=\{u,v\}} w_e \, d_e\big(\pi_{u,e}(s_u), \pi_{v,e}(s_v)\big).$$

Here s_u is one observer's state, $\pi_{u,e}$ is its readout on a shared interface, and d_e compares the two readouts in agreed terms. The weights are fixed and nonnegative. An observer revises only its own state, preserving protected evidence, and accepts a move only if the sum on its incident edges strictly falls.

Untouched edges stay unchanged, so the global sum falls by precisely the change on the touched edges. An observer can contribute to a global improvement without representing the global quantity. The mathematician uses the sum to prove something about the system; the system needs no mathematician consulting it.

An outside description of a process can be global even when nothing inside the process has global access. Requiring a coordinator because our proof mentions a total would confuse how we understand the mechanism with how the mechanism operates.

On a finite graph with finite effective state spaces, strict descent permits only finitely many accepted repairs. It does not guarantee that the last state satisfies every constraint. Nor does it guarantee that different choices of repair end together. Those are separate demands: enough repairs to reach consistency, and compatibility among alternative repair paths.

When termination and local confluence hold, Newman's (1942) lemma supplies confluence: branches from the same starting state can rejoin. Determination by evidence requires more. Distinct starting states carrying the same protected evidence must also have the same consistent public output. A private bit untouched by every repair would otherwise preserve a difference forever.

An objective answer needs to survive differences in how it is reached, and the evidence must determine the answer being claimed. Order independence alone cannot turn incomplete evidence into knowledge.

5. Inconsistency without an owner

Give three observers a binary register each, a, b, c. Their interfaces impose three requirements: $a = b$, $b = c$, and $c \neq a$.

Any one interface can be satisfied. Any two can be satisfied together. All three cannot: the first two force the opposite of the third.

The failure has no single owner. Each observer can point to a locally reasonable requirement. Yet there is no world of these three bits in which the complete specification holds. Repair may move the discrepancy around; it cannot remove an inconsistency in what it has been asked to satisfy.

Change the third requirement to $c = a$. There are two labelled solutions, 000 and 111. If flipping all bits merely changes notation, they describe one public state. If their labels have observational meaning, two possibilities remain. Protect $a = 0$ as evidence and only 000 survives.

Consistency excludes impossible candidates. Evidence can remove alternatives. Neither operation, by itself, proves physical instantiation.

6. Computation, clocks, and change

A program encourages a sequential picture: provide an input, execute instructions, obtain an output. Constraint computation separates the relation defining the answer from the procedure finding it.

Take a register relation $z = x \,\mathrm{XOR}\, y$, where the output is one exactly when the inputs differ. Fix x and y and it computes z. Fix x and z and the same relation determines y. Fix only $z = 1$ and it leaves two possible pairs. Ask for $x = y$ while protecting $z = 1$ and it reports an obstruction.

The relation has no intrinsic input end. The question acquires a direction from what the questioner holds fixed.

Local records and repairs can implement these constraints, provided permitted moves reach compatible states and preserve protected data. A solver's sequence specifies order; a physical clock can additionally measure the duration of its execution.

The word "time" needs the same care. Temporal order concerns which events precede which; physical duration adds how much time elapses along a path. A clock is a system furnishing readings. Its readings measure duration only through a justified calibration. Neither kind of time is the closure equation itself.

Five quantities must be kept apart. A repair index is a position in a solver's sequence; different schedules can take different numbers of steps to the same result. A logical clock numbers events respecting causal precedence and supplies no seconds. A physical clock is a process with countable changes, calibrated to measure proper time along its path. A time coordinate labels events in a description and is no clock. A simulator's host clock and the model's internal clock readings need not agree in rate.

Order alone fixes no duration: multiplying every logical reading by ten preserves precedence.[1] Likewise, a slower simulator can preserve all internal clock comparisons if it preserves the model's complete dynamics. A repair counter cannot silently substitute for either clock.

[1] Lamport's (1978) logical clocks make this separation precise.

Timelessness means that the complete dependency structure needs no external time to become a history. It does not remove the history's internal changes. An observer's clock is one process within that history.

A fundamental time could order events; that would not itself explain why the ordered world exists. Requiring an earlier activation of time would merely repeat the question. Closure avoids that requirement. This is its explanatory advantage, without a proof that every theory with fundamental time is inconsistent.

A result can then become a record that constrains another computation, so that the system uses a description through its own activity. An explanation of everything must include systems that use explanations.

7. The closure condition

Let $\mathfrak{U}$ denote a candidate inhabited structure. Let $D(\mathfrak{U})$ denote an account of its architecture recoverable through records within it. Let R denote an admissible realization of that account. The closure condition is

$$R(D(\mathfrak{U})) \;\cong\; \mathfrak{U}.$$

The symbol $\cong$ means agreement in the relevant observable and interventional structure. The observations used to recover the description, and the effects of actions available within the realization, must fit the world from which the account was recovered.

The equation does not picture an external computer making a smaller universe inside a larger one. Here realization means that the specified state dependencies actually obtain, including their responses to admitted interventions. It identifies the construction-side and inhabited-side readings as readings of one system. This identification is the closure hypothesis.

A recorded film can reproduce an experiment's outcomes along one history. It cannot reproduce the experiment if changing the apparatus leaves the film unchanged. A faithful realization must preserve the dependencies through which a question can enter and a consequence can be checked.

Nor does one finite observer have to contain a lossless copy of everything. Records and models can be distributed, and an architecture can be specified without enumerating every state it permits.

Self-reference earns nothing unless the return condition restricts the possibilities. On the real numbers, $T(x) = x + 1$ has no fixed point, $T(x) = x$ has every point fixed, and $T(x) = (x + 2)/2$ has exactly one. Writing $x = T(x)$ is the beginning of a question.

Lawvere's (1969) theorem gives a general account of certain diagonal fixed-point constructions under a representation hypothesis. Unrestricted self-representation fails in ordinary sets with nontrivial outputs, by Cantor's argument. Useful restricted settings must justify their admissible maps and closure properties. It would be a mistake to infer a universe from the mere presence of self-reading.

What the equation supplies here is the form of a proposed explanation. The candidate has to return its own world when its internally recoverable description is realized. Its observations cannot demand one architecture while its realization supplies another.

Nozick (1981) proposed that a fundamental principle might be self-subsuming, explaining its own holding as one of its instances. The closure condition is a self-subsuming requirement of that kind,

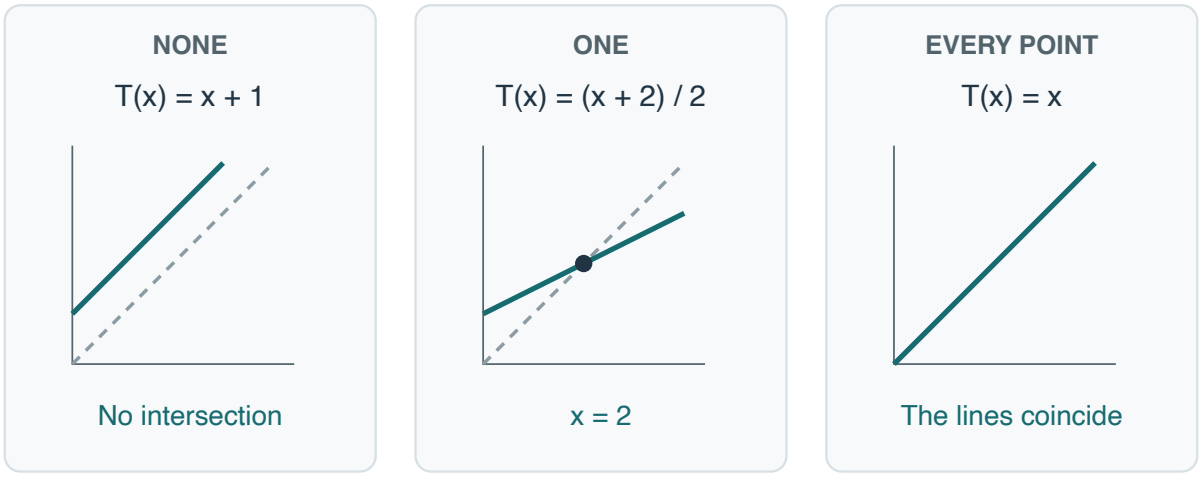


Figure 2: The same fixed-point notation admits no solution, every solution, or one solution. The explanatory work belongs to the specified map and its constraints.

with a further demand: the subsumption must be inhabited, carried out through records and agents inside the structure rather than stated about it from outside.

The question is therefore inside the answer in a specific way. The records that pose the reconstruction problem, the agents doing the reconstruction, and the activity that realizes it are constituents of the object being reconstructed. We have removed the external questioner without removing the question.

8. The meaning of "nothing"

A closure account seeks laws that follow from relations which must fit together. A quantity recovered independently from two descriptions of the same interface cannot take incompatible values. Their agreement may remove an apparent free parameter. The question changes from "who set this dial?" to "was there an independent dial?"

The architecture and both physical identifications need justification. An equation adjusted to a known answer merely encodes it, and a unique solution within one architecture need not exclude every other architecture.

Could the starting point be "nothing": a perfectly symmetric state, or a uniformly random computation?

Those suggestions describe different conditions. **Absolute absence** supplies no states, law, machine, or probability measure. A computation supplies all the structure needed to count as that computation, even when its output says "nothing." A blank tape exists; the emptiness it represents does not turn into an object by being represented.

Perfect symmetry means invariance under specified transformations. It need not mean absence of relations. Two unlabelled positions can stand in a definite relation without either being privileged. Only if every admissible state is observationally equivalent is there just one distinguishable public possibility.

Uniform randomness leaves an outcome unspecified, given a stated set of alternatives. For N equally likely alternatives, Shannon's (1948) entropy is $H = \log_2 N$, its maximum. A simple rule can specify that distribution without specifying its sampled outcome. Simplicity of the rule, uncertainty about the outcome, and absence of existence are three different things.

Here is the useful surprise. Let two bits have joint probabilities

$$P(0,0) = P(1,1) = \tfrac{1}{2}.$$

Neither bit has a preferred value. Simultaneously exchanging zero and one preserves the distribution. Yet their relation is definite: they agree. An observer holding either bit can predict the other. The symmetry survives while the relation carries information. Compare independent fair bits: the individual distributions are identical, but one tells us nothing about the other.

This makes a defensible sense of "a nothing": no preferred absolute label, while comparisons support definite relative content. The shared bit is a minimal correlation; it is no observer. Records, their causal use, and the conditions for stable public agreement must be supplied.

No choice appears by algebraic magic. A deterministic rule that respects a symmetry preserves a perfectly symmetric initial state: if $gx = x$ and $T(gx) = gT(x)$, then $gT(x) = T(x)$. An asymmetric outcome needs asymmetry, nonunique solutions with a selection rule, or stochastic selection. A symmetric distribution can have asymmetric individual outcomes.

The proposed origin is therefore logical rather than an earlier event: a world can have no privileged absolute description while possessing the relations through which its parts distinguish anything at all. This does not derive existence from absolute absence. It asks whether the demand for something *in addition to* the complete relational structure is itself misplaced.

9. Description and realization

A novel describes a person opening a door. The sentence exists; a person need not open a door. Description alone therefore cannot be a sufficient condition for the existence of its subject.

Mathematics supplies a sharper claim. Gödel's (1930) completeness theorem for classical first-order logic says that a consistent theory has a mathematical model. This concerns satisfaction of its sentences, without guaranteeing the intended model, uniqueness, or a physical world. Gödel's (1931) incompleteness results concern a different limit: sufficiently strong, effectively axiomatized arithmetic cannot both prove only truths and settle every sentence. An internally describable world need not be one whose inhabitants can prove every truth about it.

The useful distinction is between representing a dependency and realizing it. A sentence can say that a memory changes a decision. In a self-reading system, an actual retained state must help determine the continuation. The relevant connection must survive controlled changes to that state. Merely writing "because" between two events supplies no causal channel.

Consider a description with states $x \in X$ and update rules T_a, where a specifies an allowed operation. A proposed realization has actual states $s \in S$, corresponding operations F_a, and a fixed readout $p : S \to X$. Require the readout to cover the admitted described states and satisfy

$$p(F_a(s)) = T_a(p(s))$$

for every admitted state and operation. Records and ports must have identified counterparts. The operations include the relevant interventions as well as the updates that occurred on one recorded run. Their causal implementation must be established; choosing new labels after each outcome cannot count as a fixed readout.

Finite realization lemma. If this condition holds for every step, it holds for every finite admitted sequence:

$$p(F_{a_n} \cdots F_{a_1}(s)) = T_{a_n} \cdots T_{a_1}(p(s)).$$

The proof is induction. The one-step condition moves p past the last update; applying it repeatedly moves p past the whole sequence. Consequently, a described difference between two intervention outcomes must also appear in the realized readouts. The implementation carries the tested causal organization, including what would happen under the admitted alternatives.

A playback can fail this condition while matching every frame of one history. Suppose a bit is meant to toggle. After zero, both a genuine toggle and a recording can display one. Set the retained bit to one instead. The toggle produces zero; a playback that ignores the intervention displays its predetermined one.

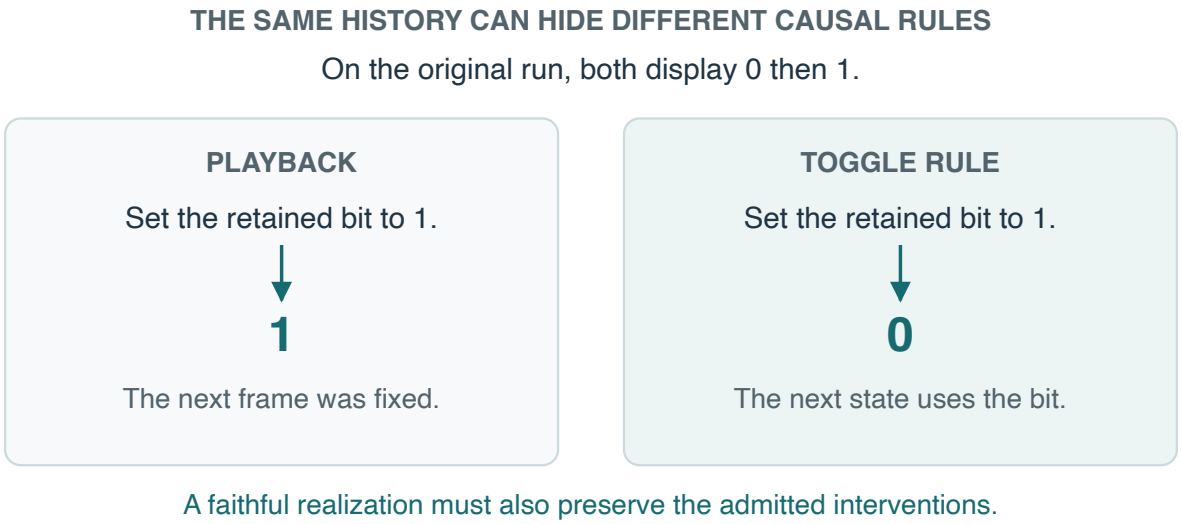


Figure 3: Matching one history does not establish the described causal rule. Changing the retained bit tests whether the next state depends on it. The difference concerns the specified state and intervention mapping.

A printed transition table fails as a running implementation when its entries do not govern corresponding state changes. Put it in a suitable interpreter and the combined system may pass. Eligibility concerns the organization supplied.

The ontological hypothesis can then be stated: **where a kind of thing is constituted by a causal organization, a realization of that organization is an instance of that kind.** The lemma checks fidelity; the constitutive hypothesis says which properties that fidelity is enough to instantiate. Chalmers's (2011) account of computational implementation develops a related distinction between a formal specification and its causal realization.

The qualification matters. A simulated flood can have actual computational consequences without putting water on the host computer's floor. A causal graph needs further geometric conditions

to constitute a spacetime. Consciousness requires the experiencing organization specified by the phenomenal hypothesis, including its records, self-reading, and ownership. A convincing display does not establish those properties.

A physical simulator gives the lemma an actual implementing system. A timeless universe raises the harder question: what realizes the entire structure when there is no external machine?

The proposed answer is **intrinsic realization**: the complete structure's lawful dependencies constitute its operation. States, records, read/write relations, and responses to admissible variations belong within it. A complete history contains change without needing a further time in which that history is played. A record of that history can be inert; the structure specified includes the dependencies through which its inhabitants act.

Here is the ontological commitment: an inhabited, causally complete closure is actual without an external act of actualization. The finite lemma does not prove this, since it assumes an actual implementation. The hypothesis extends the explanatory role of causal organization to the totality, where borrowing an external implementing machine would leave the original question unanswered. It is a restricted proposal in the direction of Tegmark's (2008) mathematical-universe hypothesis, focused on internally reconstructing observer worlds.

Why adopt it? Every test of existence available to an inhabitant is an encounter within the structure. Every clock tick, injury, correction, and discovery has consequences there. An added stamp of actuality would change none of those relations. The proposal identifies reality with the complete organization responsible for them. A philosopher can instead maintain that instantiation is an additional primitive fact; the disagreement is ontological.

"Description is existence" thus concerns the fully specified qualifying structure, rather than the ink specifying it. Writing the equation does not create the universe. On this hypothesis, its inhabitants do not wait for us to write it. A novel that actually specified such a complete structure would qualify; ordinary fiction leaves that organization unsupplied.

This is the force of the mathematical insight: the structure need not first exist as a possibility and then acquire a second property called being switched on. Its internal operation constitutes its actuality. If several closures qualify, the hypothesis admits several worlds; predicting typical observations would require a justified measure. Self-consistency alone has not selected our world.

10. Made and found

Someone invents a game and then discovers a winning strategy they did not invent. Choosing a rule and choosing its consequences are different acts.

The strange loop takes this relation further. Its inhabitants can recover the architecture of their world from its records and participate in realizing that architecture through their activity. What is discovered constrains what can be built. What is built is included in the structure from which the discovery is made.

The usual dilemma demands that one side have absolute priority: either the world exists fully before the knowers, who merely discover it, or the knowers create the world. The dilemma borrows a chronological order outside the total structure.

Inside the world, particular events have their ordinary order. Observers gather evidence, infer rules, and construct something later. The closure equation itself has no earlier stage at which one half

has to wait for the other. It specifies their mutual fidelity in a completed object.

Wheeler (1983) pictured a participatory universe as a self-excited circuit, in which observers bring about the world that gives rise to them. The closure condition keeps the circuit and removes the temporal picture. Nothing observes before there is a world to observe; the loop is a condition of mutual fidelity on the whole.

The pair of equations from section 3 showed the elementary form. We can calculate x from y or y from x. Neither calculation makes the answer arbitrary. Neither demands that one quantity existed first. The relations fix the pair together.

In the inhabited loop, the discovery is real because the consequences resist preference. The construction is real because the agents' activity is part of how the specified organization is realized. Knowing and making are coupled roles within one world.

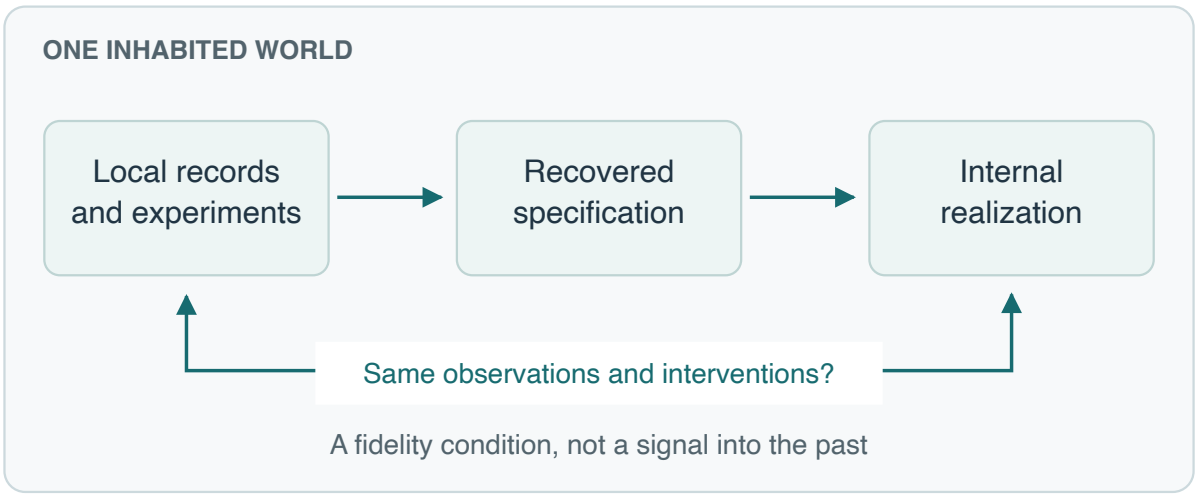


Figure 4: The closure hypothesis asks the inhabited world, an internally recovered specification, and its internal realization to agree on observations and interventions. The loop expresses mutual fidelity rather than a signal sent into the past.

What dissolves is the demand that discovery exclude participation. A musician discovers what the score requires by playing it, and the performance contributes something actual without choosing the musical relations. In the loop proposed here, the score's own recovery and realization belong to what is being performed.

The world is found in the constraints and made in their enactment. There is no contradiction until one insists that a single verb must replace both.

11. Determination and agency

The same mistake returns as a conflict between fate and freedom. If the whole world is a fixed structure, what could an inhabitant possibly decide?

Begin with the smaller theorem. Suppose a region has a unique consistent public completion for each admitted boundary record b. Write it as $N(b)$. An agent acts on the region by helping determine the record it receives.

Uniqueness says that a specified input determines an answer. It does not say that $N(b) = N(b')$ for every pair of inputs. An agent who changes the input can change the problem's solution. The inference from a determined response to an irrelevant decision has silently removed the decision from the arguments of the function.

An action can be determined through reasons and nonetheless make the difference those reasons concern. The bridge responds to its load by definite mechanics; deciding whether to place a load on it can remain consequential. The mechanics carries the decision's effect.

Within a full history, there need not be another unfinished universe waiting for us to fill in a blank. The act of deliberation is itself one of the events constituting the history. Its consideration of alternatives, sensitivity to evidence, and eventual commitment are how this part of the answer gets determined.

This dissolves the competition between a complete structure and local authorship. The structure is complete with the authorship in it. To insist that it would have the same character without the authorship is to propose a different dependency structure in place of explaining the original one.

It does not settle every theory of free will. Someone requiring an uncaused act will reject this account. The freedom defended here is a situated capacity to represent alternatives, revise commitments for reasons, and make interventions that change what follows. Those capacities distinguish an agent from a passive conduit even when both belong to a lawful world.

A person may be a small part of the world while being the place where some consequential condition is set.

12. The universe as a resolved question

A question normally seems distinct from its answer. "What is the square root of four?" is a sentence; two is a number.

Constraint computation revealed another relation. A question specifies what must fit. An answer is a state in which it fits. In a physical implementation, the question is embodied in the constraints, and answering is the activity of finding or realizing a compatible state.

Apply that form to the closure proposal. What is a universe like whose records disclose an architecture that its own activity consistently realizes? The unknown includes the records, the recovering observers, the architecture, and the realization. A satisfactory answer cannot leave any of them outside.

The universe is then the self-answering question in this precise sense. Its relations embody the conditions to be satisfied; its inhabited structure is their satisfaction; and some of its parts can make that relation explicit in a description that belongs to the same structure.

The question's physical form is the network of constraints. Its explicit formulation arrives in the record history of an observer, who can discover what the structure has been enacting without standing outside it.

This is also why observers matter to the proposal beyond being creatures we happen to like. A merely consistent abstract model need not contain any account of itself. Inhabited self-reconstruction is a stronger requirement. It asks for parts through which the world's organization can become a usable description and through which that description can enter its realization.

The extra requirement may exclude candidates that consistency alone permits. It cannot be assumed to select exactly one, and its physical implementation must be established. Its significance is that the world's being intelligible from within has entered the account of what sort of world it is.

We are accustomed to treating intelligibility as a happy accident: laws happen to exist, and much later some beings happen to discover them. The strange-loop hypothesis joins those facts. The capacities for recovery and realization are among the relations that close the structure.

13. Conclusion: the margin of the page

Return to the final explanation in your hands. The ink in the margin belongs to its subject matter. So does the alteration in your understanding, and so does the experiment you decide to build.

The explanation is being read by one of the processes it explains. That reading can change what the process does. Where the account is realized, the discovered structure becomes an effective condition of its own further activity. Its being a description does not make the activity less actual; its words matter through the relations they enter.

Discovery is the answer's resistance to arbitrary preference. Invention is the activity through which parts of it acquire determinate form. The whole structure includes the decisions; the decisions participate in determining the whole structure. None of these roles requires a first moment outside the world.

On the account defended here, this is why the question of existence has the shape of a strange loop. A universe needs no external maker when its realization and its internally recoverable account constitute one inhabited closure. The unknown on both sides of the condition is the same world.

No clock ticks outside the equation to make its world begin. The clocks, beginnings, discoveries, and decisions are among the relations it specifies. The mathematical insight becomes a claim about what constitutes their reality: the complete inhabited structure is its own realization.

"What would a universe that consistently creates itself be like?" The answer includes the systems capable of asking, the constraints their answers must respect, and the actions through which those constraints are realized.

The universe is that question answered. And the reader who finally sees the loop is one of the places where the answer has become able to state its own question.

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